Separating forcing and low-frequency feeding in a signed Navier–Stokes tail budget

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Abstract

For a smooth finite-energy forced Navier–Stokes evolution, we bound the absolute time integrals of direct force work and low-frequency feeding in the high-frequency enstrophy balance. The bounds use kinetic energy, total energy dissipation, a fixed cutoff, and derivatives of the force; they spend none of the high-frequency viscous dissipation. Consequently, for the forced singularities constructed in [1], the accumulated tail self-stretching minus even the full tail dissipation is unbounded at every fixed cutoff. The full-field obstruction is checked in Lean. The Fourier-tail extension below is an analytic proof, not a formalized projector theorem. We also bound the forcing derivative of a quartic caloric stretching functional using only energy-level velocity controls and smooth force norms. Its signed nonlinear renewal must therefore fail a uniform absorption budget on the same forced construction. No unforced regularity estimate is obtained.

1 An exact split and finite allowances

Let u, p, f solve

$$\partial_t u + (u\cdot\nabla)u = -\nabla p + \nu\Delta u + f, \qquad \nabla\cdot u = 0, \qquad \nu > 0,$$

on $\mathbb{R}^3 \times [a,T)$, where $0 < L = T - a < \infty$. Assume smoothness and, on each compact slab $[a,b] \subset [a,T)$, $u \in C^1([a,b]; H^m(\mathbb{R}^3))$ and $f, \nabla p \in C([a,b]; H^m(\mathbb{R}^3))$ for every integer $m \geq 0$. These hypotheses justify the following strong energy identities and integrations by parts. Write

$$E(t) = \|u(t)\|_2^2, \quad Y(t) = \|\nabla u(t)\|_2^2 = \|\omega(t)\|_2^2, \quad \omega = \operatorname{curl} u.$$

Suppose

$$E(t) \leq M, \quad \|f(t)\|_2 \leq F_0, \quad \|\Delta f(t)\|_2 \leq F_\Delta \qquad (a \leq t < T).$$

All constants in this note are independent of the shorter endpoint t. The kinetic-energy identity gives

$$\int_a^t Y(s)\,\mathrm{d}s \leq B, \qquad B = \frac{E(a)}{2\nu} + \frac{L\sqrt{M}F_0}{\nu}. \tag{1}$$

Indeed, $E'/2 = -\nu Y + \langle u, f\rangle$ and $E(t) \geq 0$.

Fix an angular Fourier cutoff $K \geq 0$, and set

$$u_L = \mathbf{1}_{\{|\xi|<K\}}(D)u, \quad u_H = u - u_L, \quad \omega_H = \operatorname{curl} u_H, \quad S_H = \tfrac{1}{2}(\nabla u_H + \nabla u_H^{\mathsf{T}}).$$

These are orthogonal L^2 projections commuting with derivatives. Define the physical tail quantities

$$Z_H = \tfrac{1}{2}\,\|\omega_H\|_2^2\,, \qquad D_H = \|\nabla\omega_H\|_2^2 = \|\Delta u_H\|_2^2\,, \qquad \Pi_H = \int_{\mathbb{R}^3} \omega_H \cdot S_H\omega_H \,\mathrm{d}x.$$

In particular, Π_H uses the high field in all three factors. With $\Pi = \int \omega \cdot S\omega$, introduce

$$P_L = \langle -\Delta u_L, u \times \omega\rangle, \quad P_H = \langle -\Delta u_H, u \times \omega\rangle, \quad X_H = P_H - \Pi_H, \quad F_H = \langle -\Delta u_H, f\rangle.$$

Then $P_L + P_H = \Pi$, and the exact tail balance is

$$Z_H' = \Pi_H + X_H - \nu D_H + F_H. \tag{2}$$

The filtered velocity is not assumed to solve an autonomous Navier–Stokes equation: X_H retains every lower-frequency input.

Proposition 1. *Let $c_K = (K^3/(6\pi^2))^{1/2}$. Under the preceding assumptions,*

$$\int_a^t |F_H(s)|\,\mathrm{d}s \le L\sqrt{M}F_\Delta, \tag{3}$$

$$\int_a^t |X_H(s)|\,\mathrm{d}s \le 3Kc_K\sqrt{M}B + K^2c_KM\sqrt{LB} =: C_L. \tag{4}$$

Neither bound uses a gradient supremum or tail dissipation of the velocity.

Proof. Integration by parts gives $F_H = -\langle u_H, \Delta f\rangle$, so $|F_H| \le \sqrt{M}F_\Delta$. This proves (3).

For the unitary Fourier convention, Cauchy–Schwarz on the ball gives

$$\|\nabla u_L\|_\infty \le Kc_K\sqrt{E}, \qquad \|\Delta u_L\|_\infty \le K^2c_K\sqrt{E}.$$

Here the gradient uses the Frobenius norm, and $c_K^2 = (2\pi)^{-3}|B_K|$. The vector and matrix bounds follow by Cauchy–Schwarz before taking a component sum.

For matrices define the trilinear form

$$b(A,B,C) = \sum_{i,j,k} A_{ji}B_{kj}C_{ki}, \qquad |b(A,B,C)| \le |A|_F|B|_F|C|_F.$$

The integrated stretching identity is $\Pi = -\int b(\nabla u, \nabla u, \nabla u)$, and the same identity holds for the smooth solenoidal field u_H. Put $A = \nabla u$, $H = \nabla u_H$ and $G = \nabla u_L = A - H$. The exact telescoping identity

$$b(A,A,A) - b(H,H,H) = b(G,A,A) + b(H,G,A) + b(H,H,G)$$

and $\|H\|_2 \le \|A\|_2$ imply

$$|\Pi - \Pi_H| \le 3\,\|\nabla u_L\|_\infty\, Y \le 3Kc_K\sqrt{M}\,Y. \tag{5}$$

Also

$$|P_L| \le \|\Delta u_L\|_\infty\,\|u\|_2\,\|\omega\|_2 \le K^2c_KM\sqrt{Y}.$$

Now $X_H = \Pi - \Pi_H - P_L$. Integrate these two bounds and use (1) and $\int_a^t \sqrt{Y} \le \sqrt{LB}$. □

Corollary 2. *If a finite constant A satisfies*

$$\int_a^t (\Pi_H - \nu D_H)\,\mathrm{d}s \le A \qquad (a \le t < T), \tag{6}$$

then the complete gradient energy obeys

$$Y(t) \le 2Z_H(a) + 2A + 2C_L + 2L\sqrt{M}F_\Delta + K^2 M.$$

In particular, a half-dissipation budget implies this conclusion, but the strictly smaller fraction is unnecessary for this fixed-cutoff gradient-energy implication.

Proof. Integrate (2), apply Proposition 1, and use $Y = 2Z_H + \|\nabla u_L\|_2^2 \le 2Z_H + K^2 M$. □

For an unforced solution, one can take $M = E(a)$ and $B = E(a)/(2\nu)$. Thus the low-frequency allowance is controlled by the datum, viscosity, cutoff and horizon. This does not prove (6): its uniform signed high-frequency history remains the open estimate. Allowing the full dissipation simply exposes its relation to the gradient-energy continuation criterion; it is not a new coercive estimate.

2 Application to the actual forced construction

The whole-space construction in [1] has smooth compactly supported forcing, bounded kinetic energy, zero initial velocity, and unbounded speed as $t \uparrow 1$. On slabs exhausting $[1/2, 1)$, all hypotheses above hold with fixed M, F_0, F_Δ.

The module `MettapediaNS.ForcedSignedBudget` in the Lean library applies directly to the construction's `CandidateProperties`. It proves that its gradient energy cannot have a uniform bound: such a bound would control the order-two energy via the forced viscous estimate, and the H^2 embedding would bound speed. It also proves the full-field signed-production obstruction directly, with force work bounded by $6\sqrt{E_0(u)}\sqrt{E_2(f)}$ in the doubled energy convention, where $E_j(v) = \|\nabla^j v\|_2^2$ sums all ordered derivative components. The constructor theorem is imported at the pinned revision `f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

Corollary 3. *For every such candidate, every fixed finite $K \ge 0$, and every $\beta \le 1$,*

$$\sup_{1/2 \le t < 1} \int_{1/2}^t (\Pi_H(s) - \beta\nu D_H(s))\,\mathrm{d}s = +\infty.$$

Meanwhile $\int_{1/2}^1 |X_H|$ and $\int_{1/2}^1 |F_H|$ are finite.

Proof. For $\beta = 1$, a finite upper bound would give a uniform gradient-energy bound by Corollary 2, contradicting the preceding forced-candidate theorem. For $\beta \le 1$, the accumulated quantity only increases because $D_H \ge 0$. The other assertions follow from Proposition 1. □

Thus the signed tail self-interaction is the term that defeats the budget; an infinite direct forcing-work allowance is not the cause. This conclusion applies to the complete constructed solution, without replacing its oscillatory corrections by a finite numerical model.

Forcing is still essential to that constructed history. With its zero datum and zero force, the kinetic-energy identity forces the smooth finite-energy solution to remain zero. In the construction the force is the smoothly extended momentum residual of the designed field (Section 10 of [1]). Removing it while keeping that field would change the equation. Restarting an unforced evolution from a later nonzero slice is a different problem, whose subsequent history is not settled by this argument. A proposed unforced budget must therefore use more than bounded forcing norms or bounded direct force work.

3 Forcing in a caloric stretching functional

The preceding obstruction can also test a functional that records stretching beyond the instantaneous enstrophy. Let

$$A = -\Delta, \qquad H_s = e^{-\nu s A}, \qquad R = \operatorname{curl} A^{-1} P, \qquad \mathcal{M}(v) = R\big((\operatorname{curl} v \cdot \nabla) v\big),$$

where P is the whole-space Leray projection. The multiplier R is used on a divergence: $\mathcal{M}(v) = R \operatorname{div}(v \otimes \operatorname{curl} v)$. Thus its low-frequency singularity causes no L^2 difficulty. Parseval's identity gives $\|R \operatorname{div} Q\|_2 \le \|Q\|_2$, where the tensor norm is Frobenius. For solenoidal v,

$$\|\mathcal{M}(v)\|_2 \le \|v\|_\infty \|\nabla v\|_2, \qquad \langle Av, \mathcal{M}(v)\rangle = \Pi(v). \tag{7}$$

Define the nonnegative functional

$$\mathcal{K}(u) = \frac{1}{\nu} \int_0^\infty \|\mathcal{M}(H_s u)\|_2^2 \, \mathrm{d}s. \tag{8}$$

It is finite for the smooth L^2 fields considered here: the integrand is bounded near zero and is $O(s^{-5/2})$ at infinity by heat smoothing.

The effective force in the projected velocity equation is $g = Pf$, not necessarily f itself. Suppose in addition that

$$\|g(t)\|_\infty \le Q_0, \qquad \|\operatorname{curl} g(t)\|_\infty \le Q_1 \qquad (a \le t < T).$$

These constants are finite for the smooth compactly supported force in the construction: P is bounded on every H^m, Sobolev embedding controls Pf, and $\operatorname{curl} Pf = \operatorname{curl} f$. No L^∞ boundedness of P is assumed.

Proposition 4. *Put $c_h = (8\pi)^{-3/4}$. For a fixed smooth solenoidal slice u and direction g as above, with $E = \|u\|_2^2$ and $Y = \|\nabla u\|_2^2$,*

$$|D\mathcal{K}(u)[g]| \le \frac{c_h}{\nu^2} \left(\frac{32}{3} Q_0 E^{3/4} Y^{3/4} + 16 Q_1 E^{5/4} Y^{1/4} \right). \tag{9}$$

Consequently the standing evolution satisfies, uniformly for $t < T$,

$$\int_a^t |D\mathcal{K}(u)[g]| \, \mathrm{d}s \le C_{\mathcal{K}} := \frac{c_h}{\nu^2} \left(\frac{32}{3} Q_0 M^{3/4} L^{1/4} B^{3/4} + 16 Q_1 M^{5/4} L^{3/4} B^{1/4} \right). \tag{10}$$

This payment spends no order-two velocity dissipation.

Proof. Use the auxiliary heat parameter $r = \nu s$, and write $w_r = e^{-rA} u$, $h_r = e^{-rA} g$. Differentiating both parent slots gives

$$D\mathcal{M}(w)[h] = R \operatorname{div}(h \otimes \operatorname{curl} w + w \otimes \operatorname{curl} h).$$

The multiplier bound, heat contraction and (7) imply

$$|D\mathcal{K}(u)[g]| \le \frac{2}{\nu^2} \int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2 \left(Q_0 \|\nabla w_r\|_2 + Q_1 \sqrt{E} \right) \mathrm{d}r.$$

The Gaussian kernel has L^2 norm $c_h r^{-3/4}$, and its Fourier multiplier gives

$$\|w_r\|_\infty \le c_h r^{-3/4} \sqrt{E}, \qquad \|\nabla w_r\|_2 \le \min\{\sqrt{Y}, r^{-1/2}\sqrt{E}\}.$$

When $E, Y > 0$, split each integral at $r_* = E/Y$. Direct integration yields

$$\int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2^2 \, \mathrm{d}r \le \frac{16}{3} c_h E^{3/4} Y^{3/4},$$
$$\sqrt{E} \int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2 \, \mathrm{d}r \le 8 c_h E^{5/4} Y^{1/4}.$$

These bounds also justify differentiation under the heat integral for the smooth slices. If E or Y vanishes, $u = 0$ as an L^2 whole-space field and the result is immediate. This proves (9). Finally apply Hölder in physical time, using $E \le M$, $\int Y \le B$, and the interval length at most L. □

Let $N(u) = -P((u \cdot \nabla)u)$ be the actual unforced velocity source, and set

$$\mathcal{G} = \frac{Y + \mathcal{K}}{2}, \qquad \mathcal{T} = D\mathcal{K}(u)[N(u)], \qquad \mathcal{W} = \nu \|Au\|_2^2 + \frac{1}{\nu} \|\mathcal{M}(u) - \nu Au\|_2^2 .$$

In particular $\mathcal{G} \ge Y/2$ and $\mathcal{W} \ge 0$. Heat translation in (8) gives $D\mathcal{K}(u)[-\nu Au] = -\|\mathcal{M}(u)\|_2^2/\nu$. Combine this with the actual forced enstrophy identity and expand the square in $\mathcal{W}$ to obtain the exact balance

$$2\mathcal{G}' + \mathcal{W} = \mathcal{T} + D\mathcal{K}(u)[g] + 2\langle Au, f\rangle. \tag{11}$$

Both extra force terms have absolutely bounded physical-time integrals: their sum is bounded by $C_{\mathcal{K}} + 2L\sqrt{M}F_\Delta$.

Corollary 5. *On every forced candidate of the preceding section, for every $\beta \le 1$,*

$$\sup_{1/2 \le t < 1} \int_{1/2}^t (\mathcal{T} - \beta\mathcal{W}) \, \mathrm{d}s = +\infty.$$

Proof. Integrating (11) expresses $\int(\mathcal{T} - \mathcal{W})$ as $2\mathcal{G}(t) - 2\mathcal{G}(1/2)$ minus a uniformly bounded force integral. The candidate's gradient energy has no uniform upper bound, and $\mathcal{G} \ge Y/2$. This proves the case $\beta = 1$; decreasing β only increases the integral since $\mathcal{W} \ge 0$. □

Thus the smooth force does not hide an infinite payment in the derivative of this quartic functional. Its signed nonlinear renewal is itself unbounded relative to its positive cost on the forced history. A proof of the corresponding unforced budget must exploit the zero-force evolution; estimating these explicit force terms is insufficient. The estimate does not bound $\mathcal{T}$: substituting $N(u)$ for g would require velocity-dependent supremum norms that are not controlled by the energy inequality.

4 Verification boundary

The Lean theorem `openAI_solution_no_uniform_signed_production_budget` applies to the actual construction at every positive viscosity and concerns the full-field spatial cubic integral. The sharp Fourier-tail extension and all caloric results here are analytic proofs, not Lean theorems or axioms. The companion `labs/caloric_force.py` checks the complete caloric balance by exact rational Fourier arithmetic on the periodic cube, including forces with a gradient part. These checks do not certify the whole-space estimates or a physical-time bound. Arbitrary-data unforced regularity remains open.

References

[1] OpenAI, *Finite time blowup for Navier–Stokes*, manuscript, with formalization in NavierStokesAndEuler, 2026.